

unCharted

New Delhi Metro Visualization



COM-480 Data Visualization

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1. The Idea

Our goal from the beginning was to create something that did not exist instead of reproducing another metro dashboard. During brainstorming, we looked at existing transportation visualizations and were particularly inspired by interactive projects built around the London and New York metro systems. Those projects showed that transportation data can become more than a static map: it can become an exploratory and narrative experience.

We then searched for metro systems that would be interesting from both a data and visualization perspective. We compared some of the largest metro networks worldwide and quickly noticed that the New Delhi Metro repeatedly appeared among the most extensive and most used systems globally. At the same time, unlike London, New York, or Tokyo, there were very few interactive visualizations dedicated to Delhi.

This combination made Delhi a strong candidate: the network is large enough to tell an interesting story but still underrepresented in existing visualization projects.

Once the city was selected, the next question became: what should actually be included in the website? Our objective was not to build a map with every available metric, but to create something useful and easy to explore. We wanted users to quickly understand how the network evolved while still being able to access detailed information if they were curious.

Several design decisions came from concepts discussed during the course:

- progressive disclosure instead of displaying all information at once,
- interaction as guidance rather than decoration,
- combining storytelling with free exploration,
- maintaining visual simplicity despite a large amount of data.

This led us to organize the final website around a guided introduction followed by independent exploration.

2. Design Evolution & Challenges

From Initial Sketches to Final Website

Our first milestone sketches focused mainly on three components: a geographic metro map, a temporal slider showing network expansion, and a station information panel. Those sketches defined the structure of the project but evolved considerably during implementation.

The largest change was the shift from a static dashboard mindset toward a guided user journey. Initially, users would land directly on the visualization. During testing we noticed that users often did not understand where to start or which interactions existed.

We therefore introduced a short introductory sequence at the beginning of the website that acts as a lightweight tutorial. During that tutorial we also added a storyline that can attract the attention of users. Instead of overwhelming users immediately, it progressively reveals the available interactions and creates a narrative entry point before giving full control.

This approach was strongly influenced by ideas presented during the interaction and storytelling lectures and validated through informal testing with classmates.

Challenge 1 - Building a Reliable Dataset

The most time-consuming part of the project was not visualization but data preparation. No single source contained complete and up-to-date information for the entire New Delhi Metro network.

We combined:

- official DMRC information,
- Kaggle datasets,
- Wikipedia,
- OpenStreetMap geometry.

The difficulty was that these sources often disagreed:

- station names were written differently,
- some newly opened stations were missing,
- interchange stations were represented inconsistently,
- historical opening dates did not always match.

To solve this, we manually compared stations across sources and used the official Delhi Metro information as the reference whenever conflicts appeared. This process forced us to build a cleaning and reconciliation pipeline before any visualization work could begin.

Challenge 2 - Designing a Clear User Flow

Another challenge was deciding how users should navigate the visualization. We explored several directions:

- independent tabs,
- one long scrolling page,
- a fully map-centered interface,
- guided storytelling.

Each approach changed how users interacted with the data. After multiple iterations and feedback sessions with classmates, we selected a structure that combines an introductory guided experience followed by free exploration. This decision helped preserve simplicity while keeping all information accessible.

Challenge 3 - Controlling Map Interaction

Because the website is map-based, interaction constraints became important. During early versions, users could zoom out indefinitely and move far outside Delhi, which broke immersion and made the experience confusing.

We therefore:

- limited zoom ranges and the mouse slide ranges,
- constrained panning to the Delhi region,
- adjusted transitions to remain smooth across devices.

These restrictions intentionally reduce freedom in order to improve orientation and maintain focus on the metro system itself.

Challenge 4 – Limited ridership dataset

The dataset provides only annual city-wide ridership figures, not per-station measurements. To estimate ridership at individual stations, we calculate the baseline average per station and apply a heuristic multiplier: single-line stations receive 1.0x the average, while interchange stations receive 2.5x the average to reflect their higher traffic volume.

Station estimate = (Daily Avg Ridership ÷ Total Unique Stations) x Multiplier (where Multiplier = 2.5 for interchanges, and 1.0 otherwise)

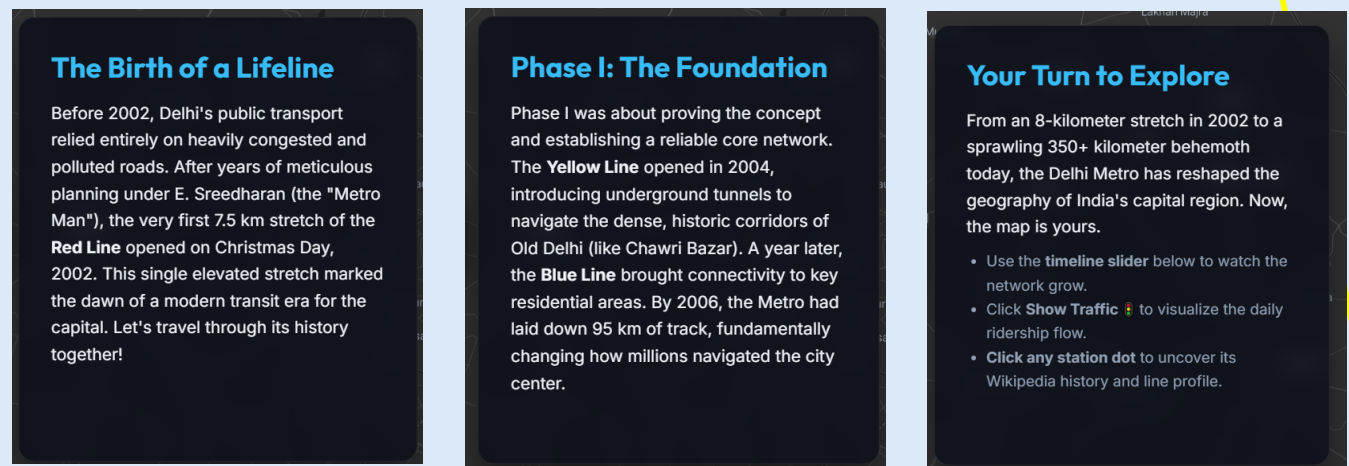
This provides a direct visual proxy for traffic at key hubs, scaling dynamically with annual ridership trends.

3. Current Website

The final website is structured as a sequence of interactive components rather than independent pages.

Introductory Story

The experience begins with a short interactive introduction. Users move through several screens using navigation controls, gradually discovering the logic of the visualization before reaching the main interface. The objective is both narrative and practical: capture attention while teaching users how to interact with the website. There is also a button Skip for the impatient ones.



Metro Network Map

The map acts as the central component. Users can navigate through the metro network, inspect individual stations, and observe geographic relationships between lines. Unlike schematic metro diagrams, the visualization preserves real geographic positioning.

Historical Timeline

A timeline slider allows users to explore how the metro expanded over time. Moving through the years progressively reveals stations and lines, showing the transformation of the network from its initial construction to the current state.

Ridership Layer

To provide more than geographic information, we introduced a ridership visualization layer. Traffic intensity is represented through glow magnitude and bubble flow

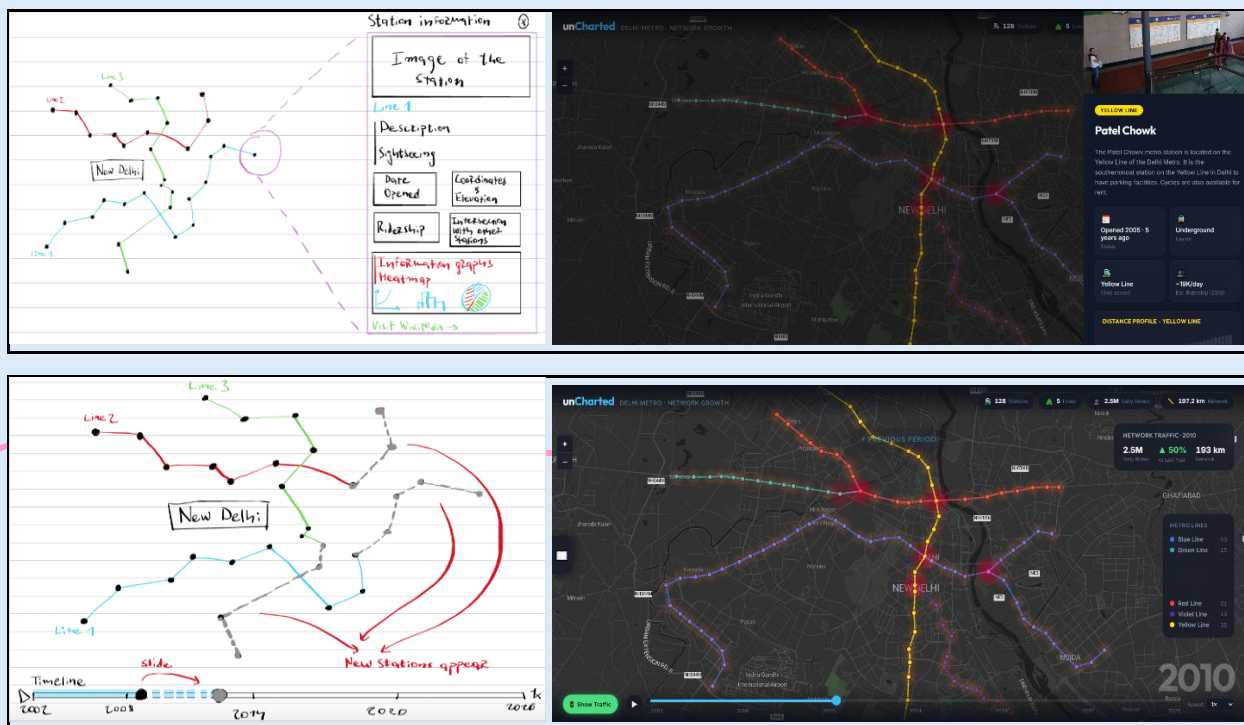
between stations, allowing users to identify busier corridors while keeping the underlying geography readable.

Network Statistics Panel

The top-right panel is non-interactive and provides high-level indicators including daily ridership evolution, total network length, number of stations, and number of active lines. These values update depending on the year to provide context while navigating.

Station Information Panel

Clicking any station opens a dedicated information panel. Users can access station metadata, opening date, ridership information, historical context, and external references through Wikipedia. This transforms the visualization from a global overview into a station-level exploration tool. The panel is also year-sensitive in a way that information changes depending on the year you landed in with the slider.



These images are comparison between the schemas and general ideas until Milestone 2 and the finished Website versions. It can be seen that we really applied what we were aiming at, and it turned out good.

4. Peer Assessment

Considering the peer assessment, we struggled to really distinguish the separate parts on which each member contributed, as we were always working together and helping each other on the parts. We successfully formed an efficient team, and considering our different backgrounds we learned a lot from each other. However, as required, we can outline the main coordinator for each task — the person in charge of planning and tracking it, while everyone participated in the work itself:

Najmeddine Abbassi

- Wikipedia scraping, data cleaning, images, descriptions etc.
- Dataset research and implementation
- Jupyter notebooks

Verter Stoilov

- Leaflet map canvas (zoom, align elements, etc.)
- Documentation, README, progress tracking
- Sketches of the front end

Malen Raychev

- Map tiles from Stadia
- D3 layer for stations, station connections, ridership density, charts
- Screencast preparation